



PhoneAI

PhoneAI - Multimodal AI Chatbot



PhoneAI

Multimodal AI Chatbot for YouTube Video QA

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7 minutes + 2 for the demo

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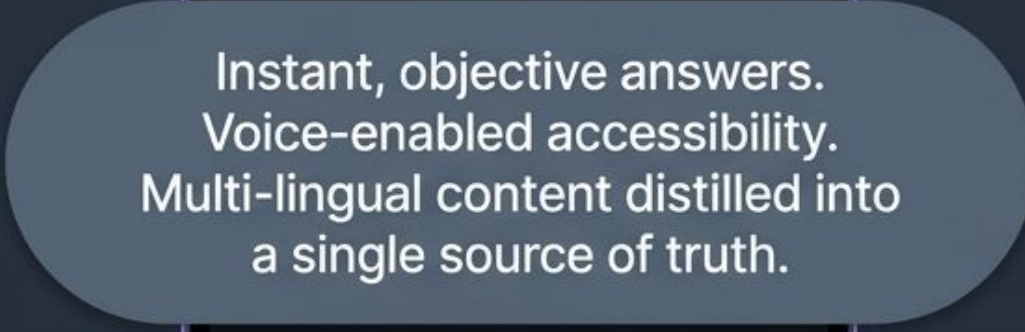
The Smartphone Buyer's Dilemma



Too many devices.
Biased marketing.
Endless 20-minute
YouTube reviews across
different languages.



Based on consensus
from 28 top reviewers,
the Galaxy S26 Ultra is
the best choice for your
camera needs.



Instant, objective answers.
Voice-enabled accessibility.
Multi-lingual content distilled into
a single source of truth.

Hook + Overview



Finding reliable tech advice hidden in hours of YouTube reviews is tedious and inaccessible for some. PhoneAI solves this by acting as the ultimate, accessible smartphone guru.



It listens to your voice



Autonomously searches through transcribed reviews from top tech YouTubers



Speaks the answer back to you



Technical Summary

- **Architecture**

Modern LangGraph React Agent architecture

- **LLM**

OpenAI gpt-4o-mini

- **Vector Database**

Pinecone (Serverless)

- **Observability**

LangSmith

- **UI Deployment**

Custom Dark Mode interface using :
Uvicorn web server, FastAPI developed with Google Antigravity IDE,
with real-time streaming

Anatomy of PhoneAI (Technical Architecture)



Dataset & Processing

• Data Source

Subtitles extracted via youtube-transcript-api from 30+ top tech channels (e.g., MKBHD, SuperSaf, Frandroid)

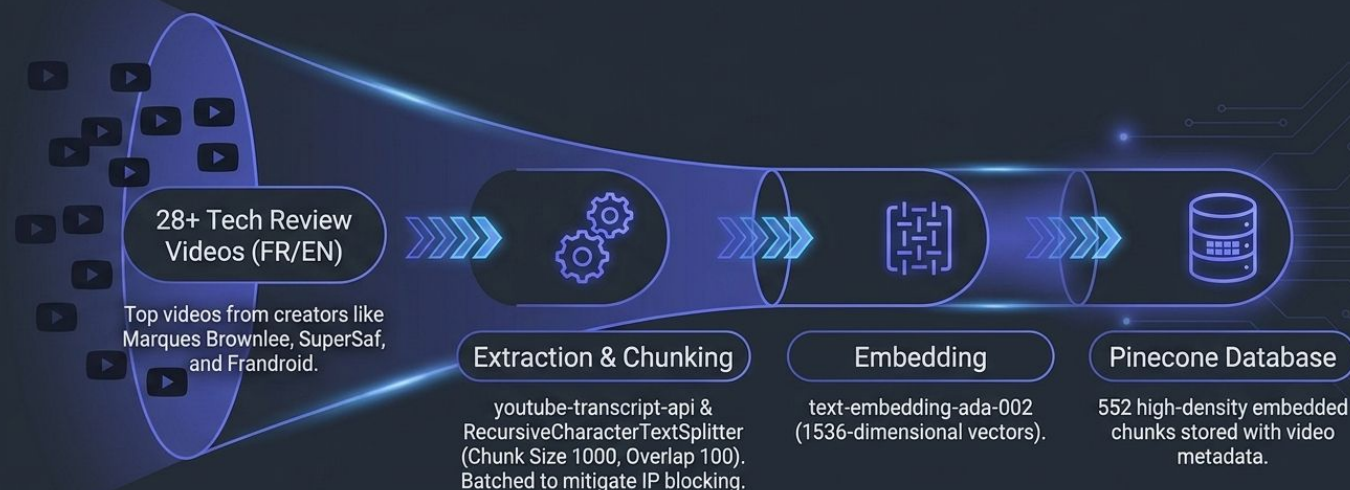
• Preprocessing & Chunking

RecursiveCharacterTextSplitter (Chunk size: 1000, Overlap: 100) to maintain semantic context

• Storage

Upserted text and metadata into a Pinecone Index (Dimension: 1536) using text-embedding-ada-002

The Dataset Pipeline: From Video to Vectors



The Q&A Agent



General Architecture

LangGraph (create_react_agent) dynamic workflow

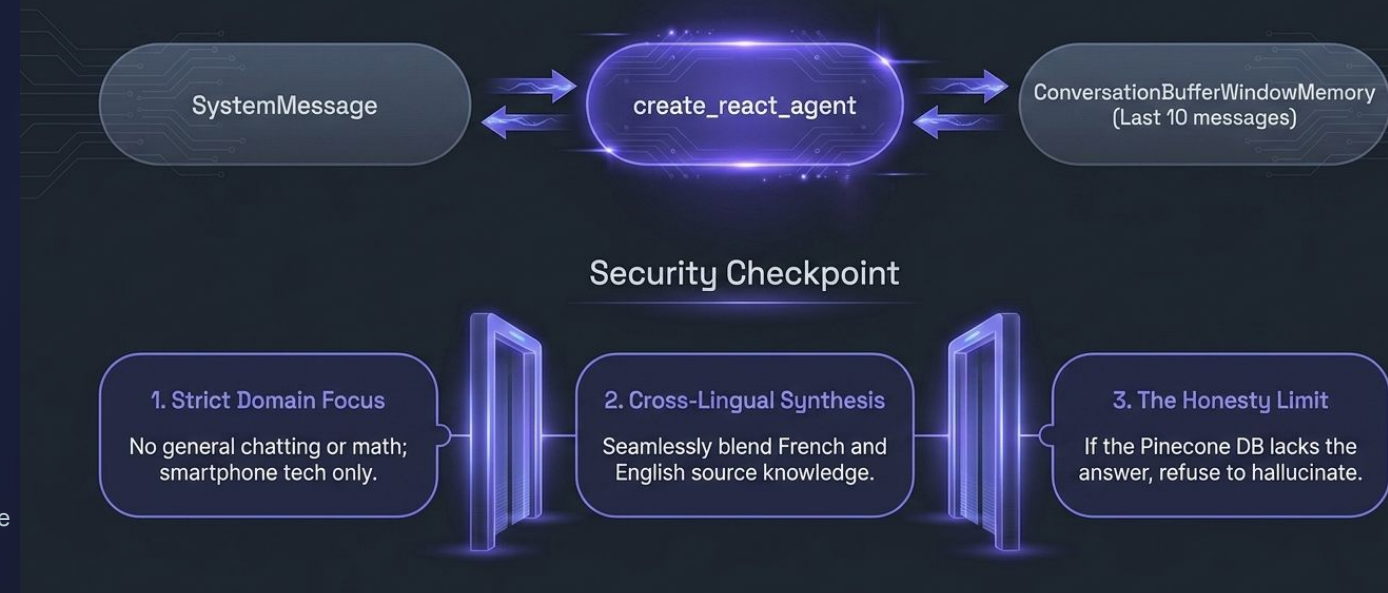
Memory

Short-term thread-level persistence, retaining the last 10 messages for contextual follow-ups

PromptTemplate

- Strict 'PhoneAI' persona
- Instructed to rely on facts
- Refuse out-of-context tech questions
- Capable of handling general conversational greetings without querying the database

The Agent Brain & Memory (LangGraph)



Tools & Multimodality

...

01

Custom Vector DB Retriever

(smartphone_tech_database)

02

Speech-to-Text

via OpenAI Whisper (whisper-1) for seamless voice input

03

Text-to-Speech

via OpenAI TTS (tts-1 alloy voice) for audio output

Highlight: The agent autonomously decides when to trigger the database tool vs. using internal knowledge

Toolset Integration & Multimodality



The Vector DB Tool

Tool Name:
smartphone_tech_database

Specs: Semantic search
configured to retrieve top 6
chunks (k=6).



Instant Speech-to-Text

Model: OpenAI Whisper-1

Function: Processes voice
input natively, bypassing text
typing.



Text-to-Speech Playback

Model: OpenAI TTS-1

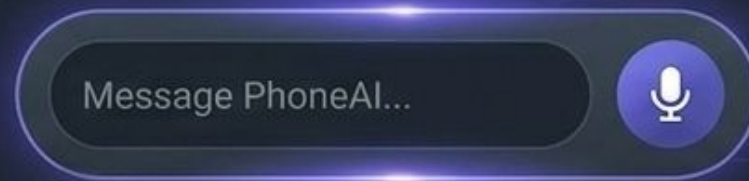
Function: Uses the Alloy voice
to read out recommendations,
enhancing accessibility.

UI & Deployment Engineering

Standard Gradio



Custom CSS Flagship UI



Premium Aesthetic

Built using `gr.themes.Soft` with custom CSS (`primary_hue='indigo'`, `neutral_hue='slate'`).

Unified Input Pill

Custom styling to merge text input and audio recording into a single, seamless component.

Token-by-Token Streaming

Iterating over `stream_mode='messages'` ensures the UI updates instantly without freezing.

Evaluation

Tools Used

- LangSmith for tracing, token usage monitoring, and latency tracking

Improvement Process

- Visual inspection of edge cases
- Iterative refinement of the PromptTemplate to eliminate hallucinations and force strict citation

Personal / Tracing / PhoneAI_Project

PhoneAI_Project ID ? Runs Threads Evaluators Automations Insights

Retention 14d Dashboard + New

Default View Last 1 day Traces Runs Add filter <>

☐	☒	Name	Input	Output	Error	Start Time	Latency
☐	☒	LangGraph	human: What are the top 3 s...	ai: The top three smartphones...		5/29/2026, 2:35:2...	7.67s
☐	☒	LangGraph	human: Feel free to ask.	ai: I'm here to assist you with ...		5/29/2026, 2:35:1...	1.18s
☐	☒	LangGraph	human: I'm searching for a fu...	ai: I'm here to help with smart...		5/29/2026, 2:34:5...	1.02s
☐	☒	LangGraph	human: What are the top 3 s...	ai: The top three smartphones...		5/29/2026, 2:29:2...	6.25s
☐	☒	LangGraph	human: What are the top 3 s...	ai: The top three smartphones...		5/29/2026, 2:28:2...	11.75s
☐	☒	LangGraph	human: Which smartphone h...	ai: For the best battery life un...		5/29/2026, 2:24:5...	4.94s
☐	☒	LangGraph	human: What are the top 3 s...	ai: The top three smartphones...		5/29/2026, 2:23:...	4.85s
☐	☒	LangGraph	human: Which smartphone d...	ai: While cats don't use smart...		5/29/2026, 2:19:4...	1.18s
☐	☒	LangGraph	human: Which smartphone c...	ai: There isn't a smartphone t...		5/29/2026, 2:19:3...	1.52s
☐	☒	LangGraph	human: Give me your most sa...	ai: The obsession with ultra-th...		5/29/2026, 2:19:3...	0.96s
☐	☒	LangGraph	human: Could you give me a ...	ai: I can't help with recipes, bu...		5/29/2026, 2:19:2...	1.01s
☐	☒	LangGraph	human: Hi PhoneAI! Quick tes...	ai: The answer to 2 multiplied ...		5/29/2026, 2:19:2...	2.64s
☐	☒	LangGraph	human: Hello PhoneAI, how a...	ai: I'm here and ready to help ...		5/29/2026, 2:19:1...	0.95s

Evaluation & Observability (LangSmith)

Trace Tracking

Used LangSmith to monitor agent node execution vs. raw tool-call JSON, preventing UI rendering errors.



Error Handling

Real-time visibility allowed rapid fixes for YouTube API IP-blocking errors during dataset ingestion.

Latency Monitoring

Tracking time-to-first-token to optimize the token-by-token streaming UX.

Conclusion & Future Scope

Project Success

Successfully engineered a multimodal, multi-lingual AI agent that bypasses YouTube fluff to deliver direct, objective, timestamp-backed answers.

Next Steps

- Implement real-time URL scraping to process brand new video reviews on demand.
- Expand the vector database to include smartwatches and tablets.



Live Demo (2 Minutes)

Demo Script Checklist:

- ✓ 1. Voice Input Test (Whisper via Microphone)
- ✓ 2. Complex Comparison (S26 Ultra vs. iPhone 17 Pro)
- ✓ 3. Audio Output Playback (OpenAI TTS)